MILITARY SPECIFICATION

CART, DENTAL ASSISTANT, MOBILE

This specification is approved for use within DLA, Defense Personnel Support Center (DM), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an assistant's cart suitable for use in the treatment of dental patients.

1.2 Classification. The assistant's cart shall be of the following types:

Type I	-	Assistant's cart without instrumentation
Type II	-	Assistant's cart with instrumentation

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-7298 - Manual, Technical, Commercial Equipment.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE/AMERICAN DENTAL ASSOCIATION (ANSI/ADA)

ANSI/ADA No. 47 Specification for Dental Units.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

UNDERWRITERS' LABORATORY, INC. (UL)

UL 544 Standard for Safety Medical and Dental Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., NY 11746, 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Road, Northbrook, IL 60062, and 1655 Scott Boulevard, Santa Clara, CA 95050).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. When materials are not specifically designated, those used shall be new, of the same or better quality as used in commercial practice, and entirely suitable for the purpose intended. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2 General requirements. The cart shall comply with the applicable requirements of American National Standards Institute/American Dental Association (ANSI/ADA) Specification No. for Dental Units, Underwriters' Laboratories, Inc. (UL) 544 Standa for Safety Medical and Dental Equipment, and additional requirement and exceptions as specified herein.

3.2.1 First article. When specified in the contract or purchas order, a sample shall be subjected to the first article inspection (see 4.3 and 6.2).

3.3 Standard commercial product. The dental cart shall, as a minimum, be in accordance with the requirements of this specificati and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited this specification but which are a part of the manufacturer's stand commercial product, shall be included in the assistant's cart being furnished. A standard commercial product is a product which has be sold or is being currently offered for sale on the commercial marke through advertisements, or manufacturer's catalogs or brochures, an represents the latest production model.

3.4 Description. The assistant's cart and instrumentation shal be designed to enhance auxiliary support at chairside and to provid unrestricted and efficient access to all instrumentation. The assistant's cart shall be furnished as Type I or Type II as specifi Requirements as specified herein are applicable to both (Type I and Type II) unless otherwise specified.

3.4.1 Type I, Assistant's Cart without Instrumentation. Type I shall be a mobile cart with a working surface and storage compartme as specified herein, (without instrumentation).

3.4.2 Type II, Assistant's Cart with Instrumentation. Type II shall consist of a mobile cart with a working surface, storage compartments, and instrumentation suitable for use by the assistant the performance of dental procedures. The cart shall be furnished with instrumentation, associated holders, hoses, controls, a utilit junction box and a vacuum solids separator as described herein. Th cart shall be suitable for central vacuum system operation unless otherwise specified (see 6.2.1). As a user specified option, an ai venturi evacuation system as described in 3.5.4.1 shall be provided when specified (see 6.2.1).

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3.5 Design, construction, and performance

3.5.1 Cart. The mobile cart shall have a work surface of no less than 225 square inches in area and storage compartments not exceeding 27 inches in depth nor 28 inches in width. A 22 inch maximum overall depth design shall be provided when specified (see 6.2.1) so as to facilitate storage under military standard counter tops. As an alternate, a primary work surface shall be augmented by an additional secondary work surface to produce an equivalent work area of 225 square inches. The working surface shall be stable and rigid so as to facilitate the performance of delicate dental procedures. The cart shall not tip when a 20 pound minimum load is placed on any edge of the work surface in the fully raised position, when tested as specified in 4.4.3.1. The work surface shall be smooth, impervious, resistant to chemicals and easy to clean. The cart shall be adjustable so as to accommodate both sit-down and stand-up operations. The minimum height shall be no greater than 30 inches so as to permit storage under military standard counter tops. The maximum height shall be no less than 37 inches. Height adjustment shall be quickly and easily performed by means of a single manual control without the use of tools. Height adjustment shall be power-assisted when specified (see 6.2.1). There shall be sufficient leg room beneath the cart for the seated dental assistant.

The cart frame shall be of all metal construction and shall resist denting, chipping, marring or scratching. The cart shall be an H-frame or U-frame design capable of withstanding a 100 pound minimum vertical load upon the work surface without collapse or structural damage, when tested as specified in 4.4.3.2. The center of gravity for top shall be aligned over the center of the base. The frame shall be plated or finished with a high quality chip and scratch resistant coating. The cart shall be freely and easily mobile by means of a minimum of4 casters. The casters shall be twin-wheel or ball type with a polished chrome body finish. The wheels shall be at least 2 inches in diameter with a stem attaching mechanism.

3.5.2 Amalgamator placement surface. The cart shall be provided with a recessed shelf, tray or other provision suitable for the placement of Dental Amalgamator, NSN 6520-00-149-0123, when specified, (see 6.2.1). The placement of the amalgamator shall be such that the height including the amalgamator shall not exceed 30 inches. An electrical receptacle shall be provided immediately adjacent to the amalgamator location. The amalgamator placement surface shall either be suitable for use by right or left handed assistants or shall be convertible into fuctionally equivalent right and left handed configurations.

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3.5.3 Storage space. Suitable space shall be provided for the storage of dental consumable supplies (needles, amalgam capsules, and anesthetic capsules). The storage space shall feature a minimum of either 4 trays, 4 bins, 4 drawers or any combination thereof. The combination, quantity and size shall be as specified (see 6.2.1). The storage space shall either be suitable for use by right or left handed assistants or shall be convertible into functionally equivalent right or left handed configurations.

3.5.4 Assistant's Instrumentation (Type II only) The cart shall be suitable for central vacuum system operation unless otherwise specified (see 6.2.1). Instrumentation shall include a three-way syringe, a high volume evacuation handpiece and saliva ejector both equipped with a rotary valve capable of on/off and variable positions so as to be capable of flow control. The vacuum lines shall be of a flexible type construction capable of withstanding 15 inches of mercury vacuum without collapse.

The cart shall be fitted with a canister type solids separator capable of removing solids from the vacuum line. The screen shall be 20 mesh, removable non-corrosive and easily accessible for cleaning. A quick disconnect water outlet with control valve shall be provided for use with accessory equipment. A rotary control valve (vacuum shutdown valve) shall be provided between the utility junction box and the solids separator so as to permit isolated shut-down of the vacuum utility for routine maintenance service, when specified (see 6.2.1). The internal bore of the valve shall not be less than the internal bore of the vaccum line so as to be non-restrictive.

All tubing shall be visually coded and shall be securely attached to barbs or fittings by positive clamps. Clamps shall be easily installed and removed. Barbs and fittings shall be constructed of durable non-corrosive materials and shall be individually replaceable.

3.5.4.1 Air venturi vacuum system. An air venturi evacuation system consisting of an air venturi vacuum device, a high volume evacuator (HVE) with a control valve, a saliva ejector with a rotary control valve; hoses, a 3-way air/water syringe and suitable holders shall be provided, when specified (see 6.2.1).

The HVE shall be capable of a flow no less than 2 cubic feet per minute at a negative pressure no less than 4 inches of mercury. The saliva ejector shall be capable of a flow of no less than 1 cubic foot per minute at a negative pressure of 15 inches of mercury. This performance shall be exacted at an air pressure not to exceed 50 psi.

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3.5.4.2 Three-way syringe. The three-way syringe shall be capable of producing a fine stream of air, water, or a mist of air and water and shall be provided with adjustments for individual flow control. Shall consist of a tip, syringe head and handle. The syringe tip shall be corrosion resistant steel and shall be autoclavable. The tip shall be rotatable for directional adjustment without loosening the attachment mechanism. The syringe tip shall have an O-ring for positive air-water separation and to prevent air leakage. The tip shall be easily removable to allow for autoclaving. The head shall be made of stainless steel, chrome plated brass, or anodized aluminum. The water and air push buttons shall be easy to remove for valve maintenance by use of a simple tool, such as a screwdriver. The handle shall be made of anodized aluminum, stainless steel, or chrome plated brass.

3.5.4.3 Syringe water anti-retraction. Syringe shall not aspirate fluids into instrument tubing so as to prevent water line contamination when measured as specified herein (see 4.4.3.3).

3.5.4.4 Control block. Control block mechanism shall be positive in action and capable of sustained operation in the presence of particulates smaller than 5 microns. The air and water supplies connected to control block shall be assembled with necessary gaskets and hardware to provide for secure instrumentation. All controls shall be easily accessible and operable and shall be clearly labelled.

3.5.4.5 Instrument holders. The assistant's instrumentation shall be securely retained in readily accessible durable plastic holders. The holders shall be designed to allow instrumentation to be easily and quickly converted into functionally equivalent right and left handed configurations either without tools or with tools supplied by the manufacturer. The right/left hand conversion shall be capable of being accomplished within 5 minutes when tested as specified in 4.4.3.4. All holders provided shall be of the same size and design.

3.5.5 Umbilical Hose (Type II only). A single umbilical shall house all tubing and wiring between the cart and junction box and shall be provided with strain relief and protection from abrasion at both connections. The umbilical hose shall be a non-collapsable tubing of sufficient length (at least 5 feet), and flexibility to permit the movement of the cart. The umbilical hose shall house all of the air, water, vacuum, drain, and electrical lines that are required to operate the cart's instrumentation. All compressed air and water tubing inside of the umbilical hose shall be a suitable plastic capable of operating safely at a pressure of 120 psi and temperature of 100 degrees F. The vacuum line shall be nominally 1/2 inch ID flexible tubing and shall not collapse under 15 inches of mercury vacuum. The umbilical hose and connections shall be designed to provide adequate protection for the internal tubing.

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3.5.6 Junction box, (Type II only). The junction box shall be an interface for connecting the assistant's cart instrumentation to the building utilities, (air, water, electricity, vacuum, and drain). The box shall be equipped with switches and valves to permit isolated shut-down from all utilities for maintenance or repair of the cart. The junction box shall not exceed 15 inches in width, 15 inches in depth and 6 inches in height and be constructed of a heavy gauge corrosion resistant metal and shall resist denting. The outside surface of box shall have a smooth finish and shall be easy to clean. The junction box shall be equipped with a lid which shall be quickly and easily removable without tools so as to allow quick and unrestricted access. The box shall be either a flush mount type or a surface mount type as specified, shall have slotted mounting holes, and shall be furnished with all fasteners required for mounting, (see 6.2.1).

The junction box shall have umbilical outlets with cover plates in each of its sides so as to permit neat and orderly installation. All tubing or wires routed through holes shall be protected from abrasion with suitable grommets.

The junction box shall be furnished with all necessary tubing and wiring to make it a functional, operating system with the cart. The shut-off valves for compressed air or water shall be corrosion resistant metal.

A manual shut-off valve, a filter and an adjustable regulator with a pressure gauge shall be provided for both air and water systems.

A water heater with adjustable thermostat shall be provided in the water system to allow adjustment of water temperature between 90° and 100°F to provide warm water for use in the syringe.

The junction box shall be fitted with easily replaceable cartridge or element type air and water filters capable of retaining particulates over 5 microns. The air and water filter shall be interchangeable. The junction box shall provide for hand-wired internal connectors to power the water heater and the umbilical-contained electrical conductors suppling the cart.

The junction box shall be equipped with a duplex electrical receptacle, when specified (see 6.2.1).

The utility junction box installation shall not require building utility stub-ups (air, water, vacuum, drain, electrical) to extend into the confines of the box, and shall not require stub-ups to conform to any template configuration other than a requirement that all stub-up ends or connection fittings be not more than 18 inches from the center of the utility junction box location. All hardware, tubing, and wiring connection of the stub-ups to the junction box shall be supplied by the cart contractor.

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3.6 Sanitization. All exposed surfaces of the cart shall be smooth and made of materials which can withstand repeated wipe down with disinfectants without deterioration of function or esthetics. No trim shall be attached such that its partial or complete loss will interfere with disinfection. All interface components that require frequent replacement such as the saliva ejector tubing, water tubing, instrument holders etc., shall be capable of being sanitized for the maintenance of proper infection control procedures.

3.7 Electrical requirements. The electrical characteristics shall be as follows, as specified (see 6.2.1).

(a)	115 ± 10% volts,	60 ± 2 Hertz
(b)	115 ± 10% volts,	50 ± 2 Hertz
(c)	230 ± 10% volts,	50 ± 2 Hertz
(d)	230 ± 10% volts,	60 ± 2 Hertz

3.8 Finish. All surfaces shall be clean and free of foreign matter, flux, or spatters of weld. The finish shall be entirely suitable for the purpose intended for the applicable component, while providing necessary protection from deteriorating effects of climate and service conditions encountered in military use. Finish on edges and surfaces shall be uniform and free from burrs, slivers, sharp edges, cracks, or other defects.

3.9 Plating. Plating shall be uniform in thickness and appearance; shall be free from blisters, peeling, visible porosity or other defects. The base metal to be plated shall be free from flaws or defects which will be detrimental to the final finish and shall be subjected to such undercoating, polishing, cleaning, dipping and plating procedures as are necessary to yield deposits with the desired function, quality, and appearance.

3.10 Workmanship. Workmanship shall be in accordance with best commercial practice. All components shall be clean, well finished, and free from defects which detract from their appearance or impair their serviceability.

3.10.1 Castings. Castings shall be sound and free from repairs, warping, or other defects that may render the casting unsound for use. Repair processes, such as welding, peeling, plugging, or filling with cold solder or metallic paste shall not be permitted.

3.10.2 Welding, brazing and soldering. All welds, brazing, and soldering shall be durable and sound in every detail. Welding shall be in accordance with the latest practices recommended by the American Welding Society. Components shall be joined without any resultant distortion and shall be free from twists or strains.

All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal.

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Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory . All aluminu parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pins holes, or cracks.

Exposed welds shall be smooth and blend with adjacent surfaces. All welds shall be cleaned, discoloration removed, and be free from slag and spatter before finishing. Welds shall be ground, only to the degree necessary to remove sharp edges or projections which may cause defects to adjacent surfaces or can be injurious to the touch.

3.10.3 Machine work. Machining shall be in accordance with best commercial practice. All bolts, machine screws, and rivet holes shal be accurate and clean. If not specified, the holes for screws and bolts shall be sized appropriately for the size screw or bolt used. All bolts, nuts, screws, and fasteners shall be completely fastened and tight. When tolerances are not specified for machine operations, tolerances shall be such as to be entirely suitable for the purpose intended.

3.10.4 Metal fabrication. Metals (sheets, plates, and tubing) shall be free from dents, folds, or distortions. Material shall be straightened by methods that will not cause injury to the metal. Operations (such as bending, shearing, or cutting) shall be done neatly and accurately. Metal forming shall be made with controlled means to insure uniformity of size and shape.

3.11 Hardware. All hardware shall be entirely suitable for the purpose intended and shall be completely compatible with the material to which it is attached or which it joins, including plated material. Any fastening device required to be loosened, tightened, or removed for repair, maintenance or adjustment shall not be swaged, peened, staked or otherwise deformed to prevent loosening of the device. All rotative fasteners shall be correctly torqued. All threaded parts shall comply with Federal Standard H28.

3.12 Identification marking. Each cart shall be marked with the following information:

CART, DENTAL ASSISTANT, MOBILE
Name of registered trademark of the manufacturer.
Contract or purchase order number.
Serial number.

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3.13 Tolerances. Where tolerances are not specified, standard commercial tolerances shall be permissible. Tolerances shall be held to a minimum where accuracy is of paramount importance for proper operation and interchangeability of component parts. Assemblies shall be precision-fitted for durable, smooth, and consistent functioning.

3.14 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.15 Service data. The contractor shall furnish, with each unit, two copies each of the technical manual which contains complete instructions for installation, operation, maintenance, and list of component parts. The manual shall comply with the requirements of MIL-M-7298. The installation manual and maintenance/repair manual shall include the following: step by step illustrated installation instructions; a list of special tools (if required); exploded drawings with part numbers; electrical and utility requirements; electrical and utility (compressed air, water, vacuum) schematic diagrams; pressure adjustment instructions and settings for air, water and vacuum, checkout procedures to ensure all operations are functioning properly; telephone number for technical assistance; and a trouble shooting chart.

The operations manual shall include the following: illustrated step by step procedures for the operation of the unit including the location and function of each control; step by step instructions for left to right conversion; pressure adjustments for air, water and vacuum; instructions for air and water adjustment for 3-way air-water syringe; instructions for the proper cleaning and disinfecting of all exposed surfaces; and a daily functional checklist.

As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and or corrections in the final manual without delaying delivery under the contract purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

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QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements. Certificates of quality may be furnished for the following: material composition, hardness, tempers, torque requirements on springs, machining and casting requirements (such as flatness, concentricity, and finish), and plating or coating.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).

2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is specified in the contract, the first article inspection shall consist of all examinations specified herein, and all testing specified herein and shall apply to the first article sample only (see 6.2). The failure of compliance with any examination or test shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 End item examination.

4.4.1.1 Visual examination. End item examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 1.0 for major defects, and an AQL of 10 for minor defects. The unit of product shall be the component of an end item or an end item, as applicable.

TABLE I. End Item Classification of Defects.

	Major	Minor
Design and Construction:		
Design and construction of unit not as specified.	X	
Hardware:		
Components, such as pins, screws, and fasteners missing, broken, stripped, cracked or deformed.	X	
Fasteners loose.		X
Finish:		
Finish not as specified.	X	
Surfaces not clean, not free of foreign matter, flux, spatters, or other defects.		X
Damage or defects on exterior or interior surfaces, such as burrs, slivers, cracks, or sharp edges which effect function or serviceability.	X	
Damage or defects on exterior or interior surfaces, such as burrs, slivers, cracks, or sharp edges which do not effect function or serviceability.		X
Plating missing (affecting function).	X	
Plating missing (not affecting function).		X
Plating not uniform in appearance.		X
Plating not free from blisters, peeling, visible porosity, or other defects.	X	
Workmanship:		
Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.	X	
Any component dented or creased.		X
Any component misplaced or not in proper alignment.	X	

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TABLE I. End Item Classification of Defects (cont.)

	Major	Minor
Workmanship:		
Any component not connected or not joined properly.		X
Fastening device requiring loosening or removal is swaged, peened, staked, or otherwise permanently fastened.	X	
Components missing.	X	
Components do not fit or mate properly.	X	
Interface fits between components not proper (too loose, too tight/binding).	X	
Components not free from defects which may impair their serviceability.	X	
Components not free from defects which do not impair their serviceability.		X
Removable components cannot be removed and replaced without difficulty.	X	
Components not properly assembled or aligned.	X	
Unit not stable on floor (not level, contains rock motion.	X	
Casters on unit do not operate or swivel freely, or bind.		X
Machine work:		
Coarse machine, tool, or die marks which affect function or serviceability.	X	
Coarse machine, tool, or die marks which do not affect function or serviceability.		X
Identification Marking:		
Identification marking missing, incomplete, not legible.		X
Identification marking incorrect or not permanent.		X
Marking not properly located.		X
Service Data:		
Service data not included.	X	

4.4.2 Dimensional examination. Dimensional examination shall be conducted for compliance with dimensions specified. Any dimension not within specification tolerance shall be a defect. The inspection level shall be S-3 with an AQL of 1.0. Unit of product shall be the components of an end item or an end item, as applicable.

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4.4.3 End item testing. End item testing shall be conducted in accordance with MIL-STD-105 and as indicated in table II, except that the acceptance number shall be zero for all sample sizes. Unit of product shall be one assistant's cart.

Table II. Sampling for Test

Component	Requirement	Test Procedure	Inspection Level
Cart stability	3.5.1	4.4.3.1	S-2
Cart vertical loading	3.5.1	4.4.3.2	S-2
Syringe water anti-retraction	3.5.4.3	4.4.3.3	S-2
Right/left hand conversion	3.5.4.5	4.4.3.4	S-2

4.4.3.1 Cart stability. A weight of 20 pounds shall be placed on each edge of the work surface in the fully raised position. The cart shall be observed for movement.

4.4.3.2 Cart vertical loading. A weight of 100 pounds shall be placed at the center of work surface. The cart shall be observed for structural damage.

4.4.3.3 Syringe water anti-retraction test. Activate the syringe spray with the tip immersed in water containing one or more drops of disclosing solution or other water soluble dye. Inactivate the syringe and allow the syringe to remain in the solution for 5 minutes. Remove the tip from the syringe and activate the syringe spray with the head over a piece of white paper. If dye is discharged, the cart is unacceptable.

4.4.3.4 Right/left hand conversion. The instruments shall be relocated from a right-handed assistant's position to a left-handed assistant's position.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level C. Each dental assistant's cart shall be packaged and cushioned in standard commercial containers of a suitable size and design that will afford adequate protection against deterioration and damage during shipment and storage.

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5.2 Packaging

5.2.1 Level C. Each packaged dental assistant's cart shall be packed in a substantial commercial exterior container (export when applicable) which will afford adequate protection to the contents during shipment from the supply source to the receiving activity. This pack shall conform to the carrier's rules and regulations.

5.3 Marking

5.3.1 Interior packages. Interior packages shall be marked to identify contents and quantity.

5.3.2 Exterior container. Each exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The assistant's cart is intended to provide enhanced auxiliary support at chairside in the treatment of dental patients.

6.2 Ordering data. Acquistion documents should specify the following:

a. Title, number and date of this specification.
b. First article, when required, (see 3.2.1),
c. Options required, (see 6.2.1).

6.2.1 The following options are listed to allow the user to choose those items required to effectively treat patients at their facility. In each category, several options are given. If no option is checked, the first listed shall be the default value.

a. Assistant's Cart type options.

(1) Cart without instruction (type I).
(2) Cart with instrumentation (type II).

b. Cart width options.

(1) Not exceeding 27 inches.
(2) Not exceeding 22 inches (under-counter version).

c. Height adjustment options.

(1) Manual type.
(2) Power assisted type.

d. Amalgamator Placement.

(1) Placement surface required.
(2) Placement surface not required.

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e. Storage space options.

(1) Trays
a. Quantity
b. Minimum size (length, width, depth)

(2) Bins
a. Quantity
b. Minimum size (length, width, depth)

(3) Drawers
a. Quantity
b. Minimum size (length, width, depth)

f. Assistant's instrumentation options.

(1) Instrumentation for use with central vacuum system.
(2) Instrumentation with air venturi vacuum system included.
(3) No assistant's instrumentation required.
(4) Additional HVE handpiece and hose required.

g. Vacuum shutdown valve options.

(1) Vacuum shutdown valve required.
(2) Vacuum shutdown valve not required.

h. Utility Junction box options.

(1) Flush mounted type.
(2) Surface mounted type.
(3) Junction box not required.

i. Electrical requirements.

(1) 115 $\pm$ 10% volts, 60 $\pm$ 2 Hertz.
(2) 115 $\pm$ 10% volts, 50 $\pm$ 2 Hertz.
(3) 230 $\pm$ 10% volts, 50 $\pm$ 2 Hertz.
(4) 230 $\pm$ 10% volts, 60 $\pm$ 2 Hertz.

j. Duplex receptacle junction box.

(1) Not required.
(2) Required.

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1960

Location: Dental-Cart

Disk-2

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mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation,
too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the
problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an
acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-42008 (DM)	CART, DENTAL ASSISTANT, MOBILE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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